A. Candies for Nephews

time limit per test: 1 second
memory limit per test: 256 megabytes

Monocarp has **three nephews**. New Year is coming, and Monocarp has n candies that he will gift to his nephews.

To ensure that none of the nephews feels left out, Monokarp wants to give each of the three nephews **the same number of candies**.

Determine the **minimum number of candies** that Monocarp needs to buy additionally so that he can give each of the three nephews the same number of candies. Note that all n candies that Monocarp initially has will be given to the nephews.

Input

The first line contains an integer t $(1 \le t \le 100)$ — the number of test cases.

Each test case consists of one line containing one integer n $(1 \le n \le 100)$ — the number of candies that Monocarp initially has.

Output

For each test case, print one integer — the minimum number of candies that Monokarp needs to buy additionally so that he can give each of the three nephews the same number of candies.

Example

input Copy

```
2
7
24
```

output Copy

```
2
0
```

Note

In the first example, Monocarp needs to buy 2 candies. After that, he will have 9 candies, and he can give each of the three nephews 3 candies.

In the second example, Monocarp does not need to buy any candies, as he initially has 24 candies, and he can give each of the three nephews 8 candies.

Educational Codeforces Round 183 (Rated for Div. 2)

Contest is running

01:55:06

Contestant

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Language: Python 3.13.2
Almost always, if you send a solution on PyPy, it works much faster

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